

Contents

Declaration	i
Certificate	ii
Acknowledgement	iii
Abstract	iv
Notations, Symbols, and Abbreviations	v
I Introduction and Literature Review	1
1.1 Project Overview	1
1.2 Problem Statement	2
1.3 Objectives	3
1.4 Scope	4
1.5 Methodology	5
1.6 Literature Review	7
1.6.1 Automated Guided Vehicles in Industry	7
1.6.2 Differential Drive Kinematics	8
1.6.3 PID Control for Line Following	9
1.6.4 ROS 2 as Robot Middleware	10

1.6.5	RFID in Warehouse Automation	11
1.6.6	Robotic Arm Manipulation with ROS 2	12
1.6.7	Industry 4.0 and Smart Warehousing	13
2	Problem Formulation	14
2.1	Core Problem Definition	14
2.2	Technical Sub-Problems	15
2.3	Design Constraints and Requirements	17
3	Modelling, Solution Methodology, and System Design	19
3.1	Overall System Architecture	19
3.2	ROS 2 Package Structure	21
3.3	AGV Mechanical Design	22
3.3.1	Physical Parameters	22
3.3.2	Real-World Chassis	23
3.4	Differential Drive Kinematics	24
3.4.1	Forward Kinematics	24
3.4.2	Inverse Kinematics	25
3.4.3	Instantaneous Centre of Rotation	26
3.4.4	Odometry Integration	26
3.4.5	Inertia Tensor	27
3.5	UR3 Robotic Arm Kinematic Model	28
3.5.1	UR3 Denavit-Hartenberg Parameters	29
3.5.2	Robotiq 2F-85 Gripper	31
3.6	Sensor Systems	31

3.6.1 Line Following Sensor Array	31
3.6.2 IMU Sensor	33
3.6.3 RFID Detection with Hysteresis	33
3.7 Control Systems Design	34
3.7.1 Line Following PD Controller	34
3.7.2 Turn Controller	35
3.7.3 Motor Velocity PID	36
3.8 Mission State Machine	37
3.9 UR3 Arm Motion Planning	38
3.9.1 MoveIt 2 Architecture	38
3.9.2 Pick-and-Place Workflow	39
3.9.3 Planner Selection Rationale	41
3.10 Hardware Architecture	41
3.10.1 Computing Platform	41
3.10.2 Power System	42
3.10.3 Simulation-to-Physical Component Conversion	43
3.10.4 GPIO Pin Mapping	44
3.11 Software Deployment Architecture	45
3.11.1 Files Unchanged for Physical Deployment	45
3.11.2 New Hardware Interface Nodes	46
3.12 GUI and Human-Machine Interface	47
4 Results and Discussions with Conclusions	49
4.1 System Performance Metrics	49

4.2	Mission Timing Analysis	51
4.3	Throughput Estimate	52
4.4	UR3 Arm Performance	52
4.5	Comparison with Commercial Systems	53
4.6	Challenges and Limitations	
54		
4.6.1	Current Limitations	54
4.6.2	Design Trade-offs	55
4.7	Future Scope	
56		
4.8	Conclusions	
57		
4.8.1	Achievements	57
4.8.2	Contributions	
59		
		63
B	Software Installation and Launch Guide	65

List of Tables

3.1 ROS 2 Package Structure	21
3.2 AGV Physical Parameters	22
3.3 UR3 Denavit-Hartenberg Parameters (Modified DH Convention)	29
3.4 Sensor Specifications	32
3.5 Mission State Machine Summary	37
3.6 AGV Computing Platform Comparison	42
3.7 AGV Power Budget	43
3.8 Simulation-to-Physical Component Conversion	44
3.9 Raspberry Pi 4 GPIO Pin Mapping	45
3.10 Code Changes for Physical Deployment	46
4.1 System Performance Metrics	49
4.2 Mission Timing Analysis (Shelf S05)	51
4.3 Comparison with Commercial AGV Systems	53
4.4 Design Trade-offs Summary	56
A.1 Complete AGV FSM Transition Table	63

List of Figures